ECE C147/C247, Winter 2025 **Midterm**
Neural Networks & Deep Learning Prof. J.C. Kao
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UCLA True Bruin academic integrity principles apply.
Open: Four pages of cheat sheet allowed, Calculator.
Closed: Book, computer, internet.
2:00pm-3:50pm PT.
Wednesday, 19 Feb 2025.

State your assumptions and reasoning.
No credit without reasoning.
Show all work on these pages.

Name: _____

Signature: _____

ID#: _____

Problem 1	_____ / 24
Problem 2	_____ / 31
Problem 3	_____ / 20
Problem 4	_____ / 25
BONUS	_____ / 7 bonus points
Total	_____ / 107 points

1. **Multiple choice and short answer.** (24 points)

(a) (3 points) You are training a softmax classifier, as we learned in class, to classify 28×28 black-and-white images into one of 7 possible classes. Assume that the weights and biases are randomly and properly initialized and the loss is the cross-entropy loss. Approximately what value should you expect the loss to be after the first forward pass?

(A) 2.3
(B) 0.81
(C) 1.9
(D) 7.0
(E) There is not enough information to answer this question.

(b) (3 points) You find that your softmax classifier is underfitting the training data and performing poorly on the validation set. Which of the following could help mitigate underfitting while also improving generalization to unseen data? Select all that apply.

(A) Add dropout during validation.
(B) Add the Frobenius norm of the weight matrix to the loss function.
(C) Remove all biases from your model.
(D) Redo the train-val-test split and assign less data as training data.
(E) None of the above.

(c) (12 points) You are developing an image classification model and considering using the Fully Connected Neural Network (FCNN). Initially, you use the SGD optimizer with a fixed learning rate to train two FCNNs with different model sizes and obtain the following training and validation accuracies:

	Training Accuracy	Validation Accuracy
Model A	0.96	0.77
Model B	0.69	0.62

(This question continues on the next page. Both of the following questions (i) and (ii) are based on this context provided in 1c.)

i. (6 points) Your teammate suggests adding Batch Normalization (BN) to Model A. In no more than three sentences, discuss whether BN can improve Model A's performance and explain why.

ii. (6 points) You observe that the training loss of Model B oscillates significantly without decreasing consistently. In no more than 3 sentences, provide a suggestion to mitigate this issue and explain why it would be effective.

(d) (6 points) Finally, you have decided to use a Convolutional Neural Network (CNN) that you learned in class to perform the image classification task described in (a). You implemented a CNN with the following architecture:

$$\texttt{conv} - \texttt{relu} - 2 \times 2\,\texttt{maxpool} - \texttt{affine}_1 - \texttt{relu} - \texttt{affine}_2 - \texttt{softmax}$$

`conv` is a convolution layer with five `5x5` filters, `stride = 1`, and `padding = 1`.

$\texttt{affine}_1$ is a fully-connect layer of dimension $\texttt{h}_1 \times 128$.

$\texttt{affine}_2$ is a fully-connect layer of dimension 128×7.

2×2 `maxpool` layer has `stride = 2`

Note: the output of the 2×2 `maxpool` is a matrix, which is reshaped into a vector of length $\texttt{h}_1$ prior to $\texttt{affine}_1$. *(This question continues on the next page.)*

Compute the value of $\texttt{h}_1$ (i.e., the number of output neurons after the 2×2 `maxpool`) and the number of parameters for the layers shown below and fill in the boxes. You may receive partial credit for correct work.

$\texttt{h}_1 =$ [ ], Number of parameters in `conv` = [ ]

Number of parameters in 2×2 `maxpool` = [ ]

Number of parameters in $\texttt{affine}_1$ = [ ]

2. **Neural network with weights and biases norm constraints** (31 points)

Consider the neural network with the following architecture

$$
\begin{aligned}
h_1 &= W_1 x^{(i)} + b_1 \\
a_1 &= \text{softplus}(h_1) \\
h_2 &= W_2 a_1 + b_2 \\
\hat{y} &= \sigma(h_2) \\
L &= \sum_{i=1}^{K} \max(0, 1 - y_i \hat{y}_i)
\end{aligned}
$$

where $x^{(i)} \in \mathbb{R}^{D_x \times 1}, W_1 \in \mathbb{R}^{D_{a_1} \times D_x}, W_2 \in \mathbb{R}^{K \times D_{a_1}}$. σ is the sigmoid activation function and y is a K-dimensional vector of 1's and 0's. y_i represents the ith element of y and $\hat{y}_i$ represents the ith element of $\hat{y}$. Softplus is a smooth approximation to the ReLU function and can be used to constrain the output to always be positive. It is an element-wise function defined as follows:

$$
\text{softplus}(x) = \frac{1}{\beta} \log(1 + e^{\beta x})
$$

In this question, you need to be very precise with the order of operations and transposes. If you miss a transpose or swap the order of a linear algebra multiplication, you will not receive points. You should also specify elementwise multiplication (i.e., the Hadamard product) with the symbol: $\odot$. You can use any gradients or backprop gradients we derived in class without proof.

(a) (3 points) Draw the computational graph for the neural network architecture described above. In the computational graph, you can write softplus as a single node (i.e., you do not have to draw the graph for $\frac{1}{\beta} \log(1 + e^{\beta x})$).

(b) (3 points) Compute $\nabla_{\hat{y}} L$. For all the following parts, you can refer to this computed gradient as $\delta_{\hat{y}}$.

(c) (3 points) Compute $\nabla_{h_2} L$. For all the following parts, you can refer to this computed gradient as δ_{h_2}. Recall the gradient $\nabla_x \sigma(x) = \sigma(x)(1 - \sigma(x))$.

(d) (3 point) Compute $\nabla_{W_2} L$. For all the following parts, you can refer to this computed gradient as δ_{W_2}.

(e) (1 point) Compute $\nabla_{b_2} L$. For all the following parts, you can refer to this computed gradient as δ_{b_2}.

(f) (3 points) Compute $\nabla_{a_1} L$. For all the following parts, you can refer to this computed gradient as δ_{a_1}.

(g) (3 points) Compute $\nabla_{h_1} L$. For all the following parts, you can refer to this computed gradient as δ_{h_1}.

(h) (3 points) Compute $\nabla_{W_1} L$. For all the following parts, you can refer to this computed gradient as δ_{W_1}.

(i) (1 point) Compute $\nabla_{b_1} L$. For all the following parts, you can refer to this computed gradient as δ_{b_1}.

(j) (8 points) Based on some apriori information, you want to impose some norm constraints on W_1, W_2 and b_1. After exploring the various norm constraints, you have decided to modify the loss function in the following way:

$$\tilde{L} = L + LSE(b_1) + \|W_1\|_\infty + \|W_2\|_F^2,$$

where Log-Sum-Exponent (LSE) of a vector $x \in \mathbb{R}^n$ is defined as follows:

$$LSE(x) = \ln\left(\sum_{i=1}^{n} e^{x_i}\right).$$

Also, recall for $A \in \mathbb{R}^{m\times n}$

$$\|A\|_\infty = \max_{1\le i\le m} \sum_{j=1}^{n} |a_{ij}|$$
$$\|A\|_F^2 = \sum_{i=1}^{m}\sum_{j=1}^{n} |a_{ij}|^2$$

Compute $\nabla_{W_1}\tilde{L}$, $\nabla_{W_2}\tilde{L}$ and $\nabla_{b_1}\tilde{L}$. **Note**: your answer should include terms $\nabla_{W_1}L$, $\nabla_{W_2}L$, and $\nabla_{b_1}L$, respectively, as these terms cannot be simplified. *(There is additional space on the next page for this question.)*

Additional space for question 2j.

3. **Elastic Net Regularization** (20 points)
In this question, we will discuss some characteristics of Elastic Net. Elastic Net Regression is an extension of linear regression that incorporates both $L1$ (Lasso) and $L2$ (Ridge) regularization penalties. The loss function is defined as:

$$\mathcal{L}(\theta) = \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\theta\|_2^2 + \lambda \left(\alpha \|\theta\|_1 + \frac{(1-\alpha)}{2} \|\theta\|_2^2 \right)$$

where n is the number of data examples, $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n \times d}$, and $\theta \in \mathbb{R}^d$ where d is the number of features.

(a) (4 points) Describe the role of α. If $\alpha \to 1$, which regularization method does Elastic Net reduce to? If $\alpha \to 0$, which regularization method does Elastic Net reduce to? Answer with no more than 3 sentences.

(b) (8 points) Briefly explain the advantages of Elastic Net compared to standalone $L1$ regression and standalone $L2$ regression. Answer this question with no more than 4 sentences.

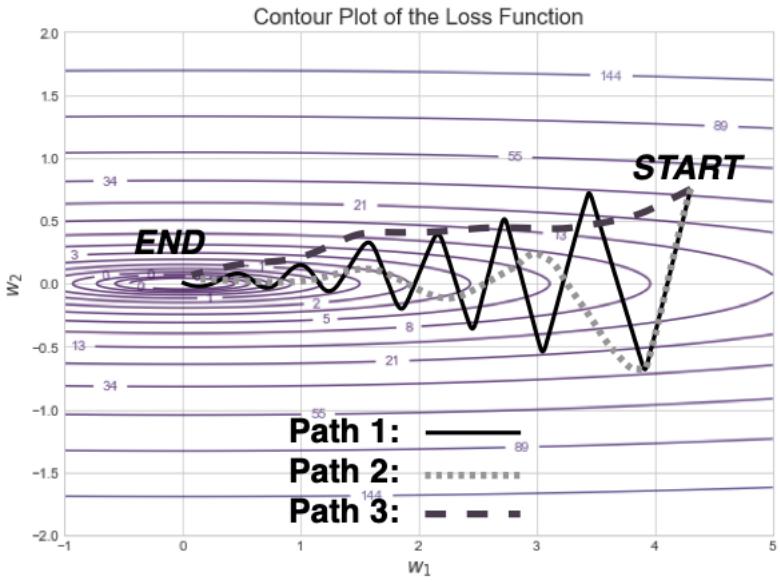
(c) (8 points) Derive the gradient of the Elastic Net loss function with respect to the parameters θ, i.e. $\nabla_\theta \mathcal{L}(\theta)$.

You may use these gradients without proof.

$$\nabla_\theta \theta^T \mathbf{w} = \mathbf{w} \qquad \text{and} \qquad \nabla_\theta \theta^T \mathbf{A}\theta = (\mathbf{A} + \mathbf{A}^T)\theta$$

4. **Optimization techniques** (25 points)

(a) (15 points) Consider a two-dimensional loss function $L(w_1, w_2)$ whose contour plot is shown in the figure below. Three distinct optimization paths are observed when minimizing L:



Match each path shown in the plot (paths i, ii and iii) to one of the following optimizers:

SGD, SGD with momentum, RMSprop

Provide a brief justification (in no more than two sentences per path) for your matching.

(b) (10 points) Given the RMSProp update equations:

$$\mathbf{v}_t = \gamma\, \mathbf{v}_{t-1} + (1-\gamma)\mathbf{g}_{t-1}^2, \quad \boldsymbol{\theta}_t = \boldsymbol{\theta}_{t-1} - \frac{\epsilon}{\sqrt{\mathbf{v}_t + \delta}} \odot \mathbf{g}_{t-1},$$

where we define $\mathbf{g}_t = \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t)$ as the gradient of L at $\boldsymbol{\theta}_t$, derive an expression for $\boldsymbol{\theta}_2$ in terms of $\boldsymbol{\theta}_0$, $\mathbf{g}_0$, and $\mathbf{g}_1$. Your derivation should clearly show how the accumulated squared gradients affect the update. **Note: For a vector $\mathbf{v}$, the operations $\mathbf{v}^2$, $\sqrt{\mathbf{v}}$, and $1/\mathbf{v}$ are all element-wise operations.**

Bonus: Generative Perspective of Softmax Classifier (7 points). **Please note that no partial credit is awarded for the bonus problem within each sub-question.**
For the classification task, a discriminative classifier directly models $P(y = c \mid \mathbf{x})$, including the Softmax classifier we covered in class:

$$P(y = c \mid \mathbf{x}) = \frac{\exp\left(\mathbf{w}_c^\top \mathbf{x} + b_c\right)}{\sum_{c'} \exp\left(\mathbf{w}_{c'}^\top \mathbf{x} + b_{c'}\right)}, \tag{1}$$

where $\mathbf{x} \in \mathcal{X}$ are the input and $y \in \mathcal{Y} = \{1, \cdots, C\}$ are the labels.

In contrast to the discriminative classifier, a generative classifier defines the class conditional distribution $P(\mathbf{x} \mid y)$ and class prior $P(y)$ to indirectly model $P(y \mid \mathbf{x})$ via specifying the joint distribution $P(\mathbf{x}, y) = P(y)P(\mathbf{x} \mid y)$.

One popular choice for a generative classifier is linear discriminant analysis (LDA). It assumes that the class conditional distribution $P(\mathbf{x} \mid y)$ follows the multivariate Gaussian distribution and the class prior $P(y)$ follows a discrete distribution:

$$P(\mathbf{x} \mid y = c) = \mathcal{N}\left(\mathbf{x} \mid \mu_c, \mathbf{\Sigma}\right), \quad P(y = c) = \frac{\beta_c}{\sum_{c'} \beta_{c'}}, \tag{2}$$

where μ_c and $\mathbf{\Sigma}$ are the mean and shared covariance of the multivariate Gaussian distribution, and β_c is the unnormalized prior for class c.

The multivariate Gaussian class conditional distribution for random variable $\mathbf{x} \in \mathbb{R}^D$ is defined as:

$$P(\mathbf{x} \mid y = c) = \frac{\exp\left(-\frac{1}{2}(\mathbf{x} - \mu_c)^{\mathrm{T}} \mathbf{\Sigma}^{-1} (\mathbf{x} - \mu_c)\right)}{\sqrt{(2\pi)^D |\mathbf{\Sigma}|}}, \tag{3}$$

where $|\mathbf{\Sigma}|$ denotes the determinant of the covariance matrix $\mathbf{\Sigma}$.

In this question, you will establish a connection between the generative LDA classifier and the discriminative Softmax classifier, providing a generative perspective on the latter.

(a) (1 point) Express the posterior distribution $P(y = c \mid \mathbf{x})$ in terms of $P(y = c)$ and $P(\mathbf{x} \mid y = c)$, where $\mathbf{x} \in \mathcal{X}$ are the input and $y \in \mathcal{Y} = \{1, \cdots, C\}$ are the labels. *Hint: Bayes' rule and marginalization.*

(b) (5 points) Using the definitions of class prior $P(y = c)$ and class conditional distribution $P(\mathbf{x} \mid y = c)$ given in equation (2) and (3), show that your expression for $P(y = c \mid \mathbf{x})$ in part (a) simplifies to

$$P(y = c \mid \mathbf{x}) = \frac{\exp\left(\mu_c^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} - \frac{1}{2}\mu_c^\top \boldsymbol{\Sigma}^{-1}\mu_c + \log \beta_c\right)}{\sum_{c'=1}^{C} \exp\left(\mu_{c'}^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} - \frac{1}{2}\mu_{c'}^\top \boldsymbol{\Sigma}^{-1}\mu_{c'} + \log \beta_{c'}\right)}. \tag{4}$$

Note: log denotes natural logarithm. *Hint 1:* $x = \exp(\log(x))$. *Hint 2:* $\exp(a + b) = \exp(a)\exp(b)$.

(c) (1 point) Show that the LDA posterior distribution in equation (4) can be converted to the form of the Softmax classifier in equation (1).

In other words, express $\mathbf{w}_c$ and b_c in terms of μ_c, $\boldsymbol{\Sigma}$, and β_c. This indicates the equivalence between LDA and Softmax classifier and provides a generative perspective for the latter.

Hint: You can still do this part without finishing part (b)

Extra page for work. If you put work here, write in that question that extra work is shown on the 'Extra Page.'